

This assignment was one of the most interesting computer vision assignments I have worked on because it combined concepts from geometry, projective transformations, deep learning, image processing, and real-world measurements. Initially, the problem appeared simple. The task was to estimate the distance of a bicycle from the Arch of Knowledge using a single image. My first thought was that object detection alone should be sufficient. However, after carefully reading the assignment and attending the lectures, I realized that the focus was not object detection but single-view metrology and projective geometry.

The first part of the assignment involved detecting the bicycle using a deep learning model. I used YOLOv8 because it is easy to use and already trained on the COCO dataset. Running the model on the provided image successfully detected the bicycle and returned its bounding box coordinates. Although obtaining the bicycle location was straightforward, I soon realized that the bounding box itself did not provide the distance to the bicycle. This was my first lesson from the assignment: object detection tells us where an object is in the image, but it does not directly tell us where the object is in the real world.

The next challenge was understanding how projective geometry can be used to estimate distances from a single image. I had previously studied concepts such as vanishing points, perspective projection, and cross-ratio in class, but this assignment forced me to apply them in practice. I spent considerable time examining the IIT Bombay image and identifying the road boundaries. By extending the road edges, I was able to estimate the vanishing point. This helped me appreciate how parallel lines in the real world appear to converge in an image due to perspective projection.

A particularly interesting part of the assignment was identifying the points A' , B' , C' , and V' . At first, I found it difficult to understand why all four points had to lie on the same line. After revisiting the theory, I realized that cross-ratio is a one-dimensional projective invariant and therefore requires collinear points. This made the blue reference line an essential part of the solution. Drawing the line and obtaining pixel coordinates using MS Paint seemed simple, but in practice it required careful attention because even a few pixels of error could affect the final distance estimate.

Another important lesson came from analyzing the bicycle bounding box. Initially, I assumed that any point inside the bounding box could be used for distance estimation. However, I discovered that the choice of reference point significantly affects the result. Since the bicycle lies on the ground plane, the most meaningful point is the ground-contact location rather than the center or top of the bounding box. This observation helped me understand how image measurements relate to physical geometry.

The second part of the assignment required taking our own photographs and repeating the process. We captured a first-person image containing a security guard and several objects of interest such as a car and a bench. We also captured a third-person image showing the image acquisition process. This part of the assignment made the problem feel much more realistic because we had to deal with our own environment instead of a carefully prepared example image.

One challenge we encountered was the lack of a known reference distance in our own scene. In the bicycle problem, the MB-to-Arch distance could be obtained from Google Maps. In our image, no such reference was readily available. This forced us to think about alternative approaches, including camera calibration and the use of camera intrinsic parameters. Although the resulting distance estimates were approximate, the exercise demonstrated how projective geometry can be combined with camera models to obtain useful measurements from a single image.

Another challenge was the manual measurement of pixel coordinates. We used MS Paint to identify important points and record their coordinates. This process highlighted how sensitive geometric

calculations can be to measurement errors. Small inaccuracies in selecting a point or estimating a vanishing point often produced noticeable differences in the final results.

Overall, this assignment significantly improved my understanding of projective geometry. Before completing the task, concepts such as vanishing points and cross-ratio seemed largely theoretical. Through the process of detecting objects, annotating images, measuring coordinates, and estimating distances, I gained a practical appreciation of how these concepts are used in computer vision applications. I also learned that deep learning and geometry complement each other: deep learning can identify objects, while geometry can reason about the spatial relationships between them.

The most valuable takeaway from this assignment was the realization that a single image contains much more information than it initially appears to. By combining object detection, perspective geometry, and careful measurement, it is possible to estimate real-world distances without requiring multiple cameras or specialized sensors. This assignment therefore provided an excellent demonstration of how mathematical concepts taught in class can be applied to solve practical computer vision problems.